

NANDHANA K S

Cybersecurity Researcher & Trainer

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PROFESSIONAL SUMMARY

Cybersecurity Researcher and Trainer with professional experience in cybersecurity education, security research, web application security, API security, security operations, and hands-on laboratory development. Currently at Offenso Hackers Academy, delivering training across multiple batches, mentoring aspiring professionals, developing practical security labs, and researching modern attack and defense techniques.

Skilled in designing interactive cybersecurity learning platforms, challenge-based environments, and security-focused web applications using Python, Flask, and React. Proficient in web application security assessment, API security testing, JWT security analysis, vulnerability assessment, and security automation. Passionate about cloud security, offensive security, and practical security education.

Proven ability to develop comprehensive training curricula, assessment modules, and lab environments that bridge the gap between theoretical knowledge and real-world application across web security, API security, cloud security, and SOC operations.

PROFESSIONAL EXPERIENCE

Cybersecurity Researcher & Trainer Sep 2025 – Present

Offenso Hackers Academy, Calicut, Kerala

- Deliver cybersecurity training programs for multiple batches covering offensive security, defensive security, and security operations, training over 100+ students across multiple cohorts.
- Conduct practical sessions on Networking Fundamentals, Linux Fundamentals, Web Security, API Security, JWT Security, Threat Intelligence, SIEM Platforms, and Security Operations.
- Design and develop hands-on cybersecurity labs, simulations, and challenge-based learning environments, creating over 50+ custom lab exercises for practical skill development.
- Develop interactive learning platforms including CTF challenge platforms and browser-based Linux terminal simulators to improve practical understanding of cybersecurity concepts.
- Mentor students through real-world attack and defense scenarios including web application testing, API security testing, reconnaissance, and vulnerability assessment.
- Conduct cybersecurity research focused on web security, API security, cloud security concepts, and security automation to inform curriculum development and training content.
- Develop technical content, training materials, practical exercises, and assessment modules for cybersecurity programs, contributing to comprehensive curriculum development.

FLAGSHIP PROJECTS

MODUS – Cybersecurity Learning Management System

Technologies: React, Flask, SQLite, JWT Authentication

Designed and developed a cybersecurity-focused learning management platform integrating student management, learning modules, examinations, CTF challenges, gamification, analytics, and browser-based Linux simulations. Serves as the primary training platform for multiple cybersecurity cohorts.

Key Features: Student Management System, Batch Management, Examination Engine, CTF Challenge Platform, Linux Terminal Simulator, XP & Gamification System, Analytics Dashboard, Notification System, Role-Based Access Control

WorkLedger – Productivity Intelligence Platform

Technologies: Flask, SQLite, Analytics Engine

Developed a productivity intelligence platform designed for trainers and researchers to track daily activities, monitor productivity, generate reports, and analyze work patterns with data-driven insights.

Key Features: Work Tracking, Task Management, Schedule Tracking, Leave Management, Batch Monitoring, Productivity Analytics, Automated Reporting

OWASP Top 10 Interactive Learning Platform

Designed and developed an interactive security learning platform covering major OWASP Top 10 vulnerabilities through simulations, practical examples, and guided learning modules for controlled educational exploitation.

JWT Learning Platform

Created an educational platform focused on JSON Web Token security concepts, authentication mechanisms, token attacks, and JWT security testing techniques.

Threat Intelligence Aggregator

Technologies: Python, VirusTotal API, AbuseIPDB API, Shodan API

Built a threat intelligence automation tool that aggregates and enriches Indicators of Compromise (IOCs) using multiple threat intelligence sources, automating collection and correlation for SOC workflows.

TECHNICAL SKILLS

Security Domains: Web Application Security, API Security, VAPT, OWASP Top 10, JWT Security, Threat Intelligence, SOC Operations, Incident Response, Digital Forensics, Network Forensics

Security Tools: Burp Suite, Nmap, Wireshark, Metasploit Framework, Nessus, OpenVAS, Nikto, Wazuh, Splunk, Autopsy, John the Ripper, Kali Linux

Cloud & Identity: Microsoft Entra ID, Microsoft Defender, Conditional Access, Identity & Access Management (IAM), Microsoft Security Fundamentals, Cloud Security Fundamentals

Development & Automation: Python, Flask, FastAPI, Streamlit, HTML, CSS, JavaScript, React, SQLite, Git, GitHub

CERTIFICATIONS

- Microsoft SC-900 – Security, Compliance & Identity Fundamentals
- Palo Alto Networks – Security Operations Fundamentals
- Cisco Networking Academy – Introduction to Cybersecurity
- EC-Council – Ethical Hacking Essentials
- EC-Council – Network Defense Essentials
- EC-Council – Digital Forensics Essentials

EDUCATION

Master of Computer Applications (MCA) – Indira Gandhi National Open University (IGNOU) – *Pursuing*

Bachelor of Science (B.Sc.) in Digital & Cyber Forensic Science – Bharathiar University – 2022–2025 – CGPA: 8.3

TEACHING & TRAINING EXPERTISE

Topics Delivered:

- Cybersecurity Fundamentals, Networking Fundamentals, Linux Fundamentals
- Web Application Security, OWASP Top 10, API Security, JWT Security
- Security Operations Center (SOC), Threat Intelligence, Wazuh SIEM, Splunk SIEM
- Active Directory Fundamentals, Active Directory Pentesting Concepts
- Android Pentesting Fundamentals, Digital Forensics Fundamentals
- Information Gathering & Reconnaissance

RESEARCH INTERESTS

- Web Application Security, API Security, Cloud Security
- Security Automation, AI in Cybersecurity
- AI Security & Prompt Injection
- Threat Intelligence, SOC Operations, Digital Forensics
- Security Education Platforms